Saman Mallahi

Address: Unit 3, No. 13, Sattarkhan St., Shiroodi Ave., Arak, Markazi, Iran
Postal Zone/ZIP Code: 3819745351
Email: MALLAHI.SAMAN@gmail.com
LinkedIn: linkedin.com/in/saman-mallahi
Tel.: +98 920-617-0257

PROFILE

Mechanical Engineer with strong background in multibody dynamics, computational modeling, optimization, and control-oriented mechanical design. Experienced in simulation-driven engineering using Python, MSC ADAMS, CFD, and FEM tools. Academic training in Space Robotics, Artificial Intelligence, and Dynamic Systems. Highly motivated to pursue advanced studies in Robotics, Intelligent Systems, and Autonomous Mechanisms.

EDUCATION

Sapienza University | Rome, ITALY *Sep. 2017 – -/---*
Master's degree in Space and Astronautical Engineering
Key Modules:

- **Space Robotics**
- **Dynamics of Flexible Multibody Systems**
- **Attitude Dynamics and Control**
- **Neural Networks**
- **Machine Learning**
- **Artificial Intelligence**

Dissertation: --

Note: Program discontinued due to COVID-19 and personal circumstances.

Islamic AZAD University| Arak, IRAN *Sep. 2010 – Jan. 2016*
Bachelor's degree in Mechanical engineering (Solid Design)
Dissertation: design of Pressure vessels

EXPERIENCE

Petro Sazeh Mihan Arak, IRAN *Dec 2021 – Nov 2023*

R&D Assistant and Designer

- Developed multibody dynamic simulations using MSC ADAMS for torque and motion optimization.
- Built Python-based numerical models for geometry analysis and nonlinear optimization (Pyomo).
- Designed actuator sizing and performance prediction tools.
- Performed CFD simulations (ANSYS Fluent, CFX) for fluid--structure interaction studies.
- Applied DFMA principles in mechanism design and system optimization.
- Developed engineering software with Qt GUI and SQLite database integration.

- Prepared technical drawings according to ASME Y14.5 and Y14.100 standards.

Projects:

- **Design of Triple offset butterfly valve**
  - Provided a Design handbook, including all the theories and calculations necessary for the design, for all possible types of seal rings and mechanisms.
  - Provided a computer code for numerical analysis of the 3D geometry of the seal to check for interference and calculate the three offsets.
  - Provided an Optimization method for Torque, Contact pressure, controllability zone, valve characteristic & friction forces, with help of **MSC ADAMS** and **Python Pyomo** library**.**
- **Development of Control Valve Sizing software**
  - Provided a code for calculation of valve CV and noise, according to **ISA 75.01.01** standard
  - Provided a code for analysis of control valve noise according to **ISA-75.01.01-2007 (IEC 60534-2-1 Mod)** standard.
  - Provided a code for calculation of Actuator sizing
  - Prepared a professional **GUI** with the help of the **QT framework** and a database of specs, of our designed valves, with **SQLite language**
- Design of a calculation tool to compute the contours and the patterns of on cages and plugs of control globe valves.
- Establishing a structure for archiving engineering documents
- Design of cryogenic ball valves and preparing test procedure for cryogenic valve test
- Design of metal seat and soft seat ball valve with fire-safe design
- Design various types of triple-offset butterfly valve with fire-safe design
- Calculation and analysis of conventional Safety valve's springs
- Design of Spiral gaskets winding device.
- Establishing a project management and task tracking system for the company with help of **OpenProject** and **Nextcloud**

Takvin Azmayesh Parseh Arak, IRAN ***Dec. 2015 – March 2016***

Research Assistant

- Research on different types of silicone oils used as heat transfer fluid and analysis of their qualities.
- Design of silicon sealing and selecting the proper material that can tolerate different temperatures.
- Design of pressure vessel made out of food-safe alloy

Azarab Industries Arak, IRAN ***Jan 2015 – Sep 2015***

Intern

- Introduction to standards and regulations for the design of pressure vessels, including ASME BPVC Section VIII and some local Standards.
- Introduction Finite Element Analysis of Pressure Vessels.

ADDITIONAL SKILLS

TECHNICAL INTERESTS

- Robotic Mechanisms Design
- Multibody Dynamic Simulation
- Motion Planning & Optimization
- Actuator Modeling
- Intelligent Mechanical Systems
- Control-Oriented Mechanical Design
- Computational Robotics

TECHNICAL SKILLS

Programming & Computational Tools

- Python (NumPy, SciPy, Pyomo, optimization libraries)
- Qt (GUI development)
- SQLite
- Git & GitHub
- Linux

Simulation & Engineering Software

- MSC ADAMS (Multibody Dynamics)
- ANSYS Fluent & CFX (CFD)
- Abaqus (FEM)
- SolidWorks (CAD & CFD)
- CATIA
- AutoCAD

Engineering Competencies

- Multibody Dynamics
- Optimization Methods
- Actuator Modeling
- CFD & FEA
- GD&T
- DFMA

LANGUAGE & TEST SCORES

- TOEFL IBT: 94
- IELTS: 6.5
- GRE: Q163 | V152 | AWA 3